

Career Planning

Input File: *standard input*

Output File: *standard output*

You’ve just scored an entry-level job at the head offices of Complete Geodesics Inc., a prestigious multinational company that provides people around the world with globes, domes, and all sorts of other round objects.

CompGeode headquarters are a towering metropolitan building H floors high, with W private offices per floor. Your seniority in the company is determined purely by what floor your office is in: the top executives get the top floor, whereas the lowliest wage-slaves are stuck on the bottom floor. Each office also has a *classiness*, an integer score that determines how jealousy-inducing it is. (Low scores indicate bare walls and concrete floors; high scores indicate gold-lined champagne fountains, exotic rugs, and a dozen work experience kids waiting on you hand and foot.)

While you’re happy to have gotten the job – you even get to choose which first-floor office you start in! – you have great ambitions. One day, you swear, you’ll be working on the top floor. CompGeode has a *flexible promotion policy*: employees can be promoted to a new office anywhere between 1 and K floors (inclusive) above their current position.

Knowing that you hate to lower your standards, you realise that no matter how juicy a promotion sounds, you won’t be able to cope with moving to an office with lower classiness than your current one. With this in mind, you wonder how many different ways you can start at a first-floor office and undergo promotions until you reach the top floor.

Input

The first line of input will contain H and W separated by a space. The second line will contain the integer K : the maximum number of floors you can move up in a single promotion.

The following H lines describe the office floors, in order from top to bottom. Each line contains W integers describing the classiness of the offices on that floor.

Output

Output should consist of a single integer: the number of ways you can start at a first-floor office and undergo promotions until you reach the top floor, given that you cannot move from an office to another office with lower classiness. Give your answer modulo 1 000 000 007.

(continued over ...)

Sample Input 1

```

3 3
2
13 17 1
1 6 13
8 2 100

```

Sample Output 1

```

10

```

Sample Input 2

```

9 1
7
90
80
70
60
50
40
30
20
10

```

Sample Output 2

```

127

```

Sample Input 3

```

2 2
2
2 2
2 2

```

Sample Output 3

```

4

```

Explanation

In the first sample case, there are four sequences going directly from the first floor to the top floor (starting from the offices with classinesses 8 or 2, and ending at the top-floor offices with classinesses 13 or 17). There are also six sequences visiting all three floors: $\{2 \rightarrow 6 \rightarrow 13, 2 \rightarrow 6 \rightarrow 17, 2 \rightarrow 13 \rightarrow 13, 2 \rightarrow 13 \rightarrow 17, 8 \rightarrow 13 \rightarrow 13, 8 \rightarrow 13 \rightarrow 17\}$.

In the second sample case, you can take any sequence of promotions up to the top *except* for going directly from the bottom floor to the top floor. This gives $2^7 - 1 = 127$ possible paths.

In the third sample case, you can start from either office in the bottom floor and end in either office in the top floor.

Subtasks & Constraints

For all cases, $2 \leq H$, $1 \leq W$ and $1 \leq K \leq H$.

- For Subtask 1 (30 points), $W, H \leq 100$, $K \leq 50$ and all classiness values are integers between 1 and 100 inclusive.
- For Subtask 2 (30 points), $W, H \leq 500$ and all classiness values are integers between 1 and 100 inclusive.
- For Subtask 3 (40 points), $W, H \leq 1000$ and all classiness values are integers between 1 and 1 000 000 inclusive.